



MATHEMATICS (EXTENSION 1)

2014 Preliminary Course Assessment Task 2

Wednesday April 9, 2014

General instructions

- Working time – 75 minutes.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.
- Commence each new question on a new page. Write on both sides of the paper.
- **All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.**

Class (please ✓)

- ☐ 11M3A – Ms Ziazaris
- ☐ 11M3B – Mr Ireland
- ☐ 11M3C – Mr Lin
- ☐ 11M3D – Mr Weiss
- ☐ 11M3E – Mr Berry
- ☐ 11M3F – Mr Lam

STUDENT NUMBER: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	6	Total	%
MARKS	15	15	11	12	15	14	82	

Question 1 (15 Marks)	Commence a NEW page.	Marks
(a) $f(x)$ is defined as		
	$f(x) = \begin{cases} x^2 & x < 2 \\ 6 - x & x > 2 \end{cases}$	
i. Evaluate $f(-1)$ and $f(6)$.		2
ii. Sketch the graph of $y = f(x)$ within the domain $-1 \leq x \leq 6$.		3
(b) If $f(x) = 5x - 7$, evaluate $f\left(\frac{7-x}{5}\right)$.		2
(c) Show that $f(x) = \frac{1}{x(x^2-1)}$ is an odd function.		2
(d) Sketch the following graphs, showing all important features:		
i. $(x-2)^2 + (y+1)^2 = 4$.		3
ii. $y = \frac{x-1}{x+4}$.		3

Question 2 (15 Marks)	Commence a NEW page.	Marks
(a) Fully factorise:		
i. $10x^2 + 11x - 6$.		1
ii. $x^3 - 9x$.		1
iii. $x^3 + x^2 + x + 1$.		2
(b) Find a value for x such that $\sqrt{18} - \sqrt{2} = \sqrt{x}$.		2
(c) Fully simplify: $\sqrt[3]{\frac{x^{5k+1}y^{4k-5}}{x^{2(k-1)}y^{-2k+1}}}$		2
(d) Solve the following for x :		
i. $(3x+1)^2 = 6x+2$.		2
ii. $ 3x-5 = 5x$.		2
iii. $\frac{5}{x-1} \geq 2$		3

Question 3 (11 Marks)	Commence a NEW page.	Marks
(a) Write 36° in exact form in radians.		1
(b) State the exact value of $\cot \frac{11\pi}{6}$.		2
(c) Fully simplify: i. $\cos(x + y) \cos y + \sin(x + y) \sin y$. ii. $\cos^4 \theta - \sin^4 \theta$.		1 2
(d) Prove, showing full working:	$\frac{\cos \theta}{1 - \sin \theta} + \frac{1 - \sin \theta}{\cos \theta} = 2 \sec \theta$	3
(e) Given $\tan 22.5^\circ = \sqrt{2} - 1$, express $\tan 67.5^\circ$ in the form $a + \sqrt{b}$.		2

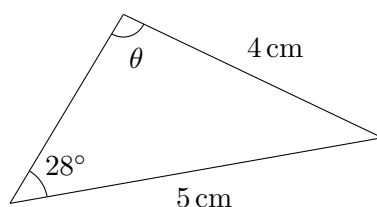
Question 4 (12 Marks)	Commence a NEW page.	Marks
(a) Given that $\sec \theta = \frac{13}{5}$ and $\tan \theta < 0$, find the exact value of the following: i. $\sin \theta$. ii. $\cos 2\theta$.		2 2
(b) Solve for θ : i. $\cos \theta = -\frac{\sqrt{3}}{2}$ for $-180^\circ \leq \theta \leq 180^\circ$. ii. $\sqrt{2} \sin 3x = -1$ for $0^\circ \leq \theta \leq 360^\circ$ iii. $2 \sin^2 \theta + 3 \cos \theta = 3$ for $0^\circ \leq \theta \leq 360^\circ$		2 3 3

Question 5 (15 Marks)

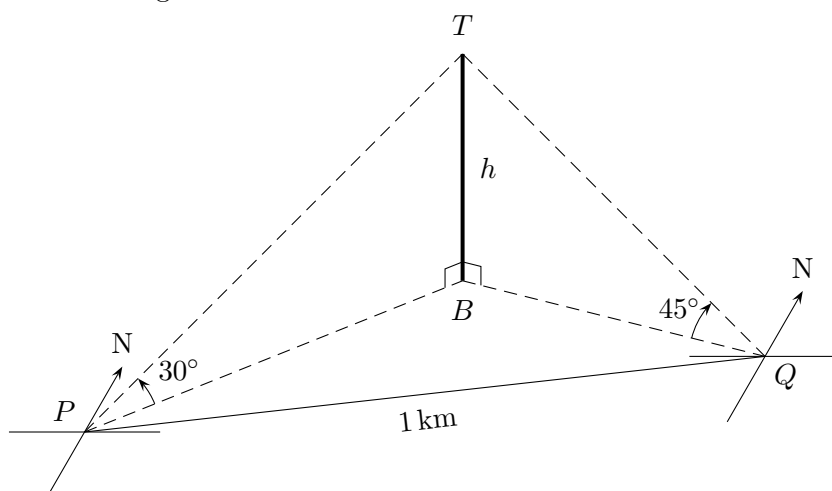
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Marks

- (a) For the function $f(x) = -\frac{5 \cos 3x}{2}$,
- State the period (in degrees) and amplitude. 2
 - Sketch the graph of $y = f(x)$ for $0^\circ \leq x \leq 360^\circ$. 3
- (b) Find the value(s) of θ : 2



- (c) The angle of elevation from a boat P to a point T at the top of a vertical cliff is 30° . The boat sails 1 km to a second point Q , from which the angle of elevation to T is 45° . Let B be the point at the base of the cliff directly below T , and let $h = BT$ be the height of the cliff in metres.



The bearings of B from P and Q are 050°T and 310°T respectively. Copy or trace the diagram into your working booklet.

- Show that $\angle PBQ = 100^\circ$. 2
- Find expressions for PB and QB in terms of h . 1
- Hence show that 2

$$h^2 = \frac{1\,000^2}{4 - 2\sqrt{3} \cos 100^\circ}$$
- Hence or otherwise, calculate the height of the cliff correct to the nearest metre. 1
- Find the area of the sea that $\triangle PBQ$ encloses. 2

Question 6 (14 Marks)

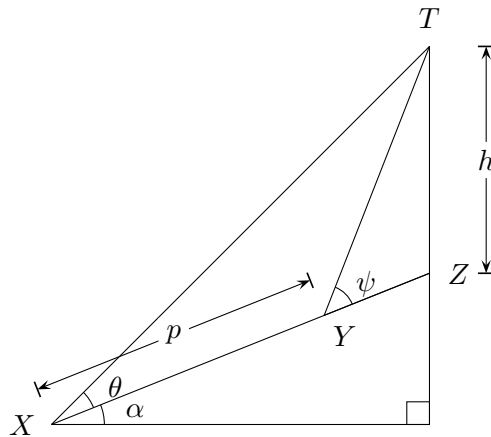
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Marks(a) Draw a neat sketch of $y = |4 - x| - x$. **3**(b) Solve simultaneously: **3**

$$\begin{cases} y = 1 - \frac{3}{x+2} \\ y = 2x \end{cases}$$

(c) Solve for $0 \leq x \leq 2\pi$, giving your solutions as exact values: **3**

$$\sin x - \sin 2x = 0$$

(d) A ramp XYZ , rises at an angle of α with $XY = p$ metres. At Z there is a building of height h metres.The angles of elevation of T from X and Y are $(\theta + \alpha)$ and $(\psi + \alpha)$ respectively.i. Show that $TY = \frac{p \sin \theta}{\sin(\psi - \theta)}$ **1**ii. Hence show that **2**

$$h = \frac{p \sin \theta \sin \psi}{\sin(\psi - \theta) \cos \alpha}$$

iii. Given that $\alpha = \theta = \frac{\pi}{6}$ and $\psi = \frac{\pi}{4}$, find the exact value of h in terms of p . **2****End of paper.**

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Suggested Solutions

Question 1 (Ireland)

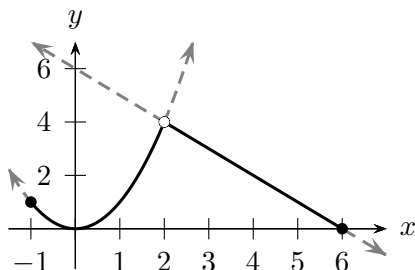
(a) i. (2 marks)

$$f(x) = \begin{cases} x^2 & x < 2 \\ 6 - x & x > 2 \end{cases}$$

$$f(1) = (-1)^2 = 1$$

$$f(6) = 6 - 6 = 0$$

ii. (3 marks)



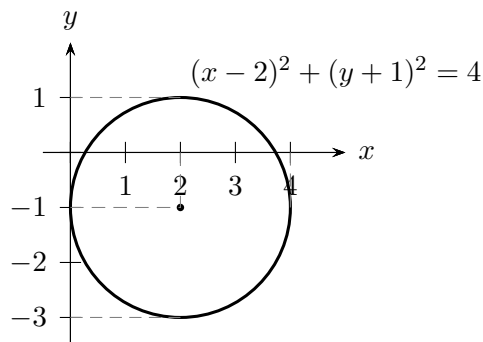
(b) (2 marks)

$$\begin{aligned} f(x) &= 5x - 7 \\ f\left(\frac{7-x}{5}\right) &= 5\left(\frac{7-x}{5}\right) - 7 \\ &= 7 - x - 7 \\ &= -x \end{aligned}$$

(c) (2 marks)

$$\begin{aligned} f(x) &= \frac{1}{x(x^2 - 1)} \\ f(-x) &= \frac{1}{(-x)((-x)^2 - 1)} \\ &= -\frac{1}{x(x^2 - 1)} \\ &= -f(x) \\ f(-x) &= -f(x) \\ \therefore f(x) &\text{ is odd} \end{aligned}$$

(d) i. (3 marks)



ii. (3 marks)

$$\begin{aligned} y &= \frac{x-1}{x+4} \\ &= \frac{x+4-5}{x+4} \\ &= 1 - \frac{5}{x+4} \end{aligned}$$

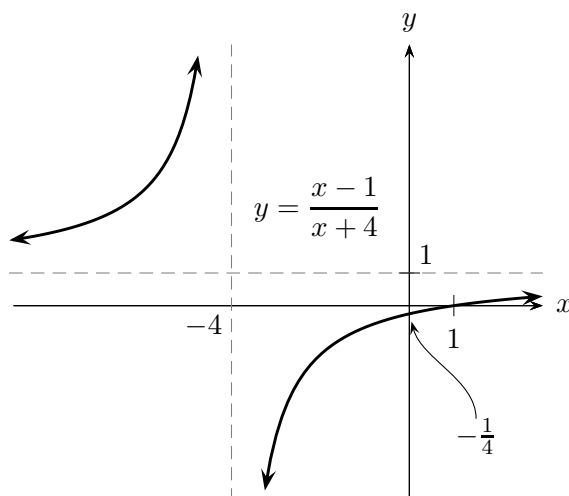
Intercepts:

• $x = 0$:

$$y = 1 - \frac{5}{0+4} = -\frac{1}{4}$$

• $y = 0$:

$$\begin{aligned} 0 &= 1 - \frac{5}{x+4} \\ \frac{5}{x+4} &= 1 \\ x+4 &= 5 \\ x &= 1 \end{aligned}$$



Question 2 (Ireland)

(a) i. (1 mark)

$$10x^2 + 11x - 6 = (5x - 2)(2x + 3)$$

ii. (1 mark)

$$\begin{aligned} x^3 - 9x &= x(x^2 - 9) \\ &= x(x - 3)(x + 3) \end{aligned}$$

iii. (2 marks)

$$\begin{aligned} x^3 + x^2 + x + 1 \\ &= x^2(x + 1) + (x + 1) \\ &= (x + 1)(x^2 + 1) \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} \sqrt{18} - \sqrt{2} &= 3\sqrt{2} - \sqrt{2} \\ &= 2\sqrt{2} \\ &= \sqrt{8} \\ \therefore x &= 8 \end{aligned}$$

(c) (2 marks)

$$\begin{aligned} &\sqrt[3]{\frac{x^{5k+1}y^{4k-5}}{x^{2(k-1)}y^{-2k+1}}} \\ &= \sqrt[3]{x^{5k+1-2(k-1)}y^{4k-5-(-2k+1)}} \\ &= \sqrt[3]{x^{3k+3}y^{6k-6}} \\ &= x^{k+1}y^{2(k-1)} \end{aligned}$$

(d) i. (2 marks)

$$\begin{aligned} (3x + 1)^2 &= 6x + 2 \\ 9x^2 + \cancel{6x} + 1 &= \cancel{6x} + 2 \\ 9x^2 &= 1 \\ x^2 &= \frac{1}{9} \\ x &= \pm \frac{1}{3} \end{aligned}$$

ii. (2 marks)

$$|3x - 5| = 5x$$

When $(3x - 5) \geq 0$,

$$\begin{aligned} 3x - 5 &= 5x \\ 2x &= -5 \\ x &= -\frac{5}{2} \end{aligned}$$

When $(3x - 5) < 0$,

$$\begin{aligned} -(3x - 5) &= 5x \\ -3x + 5 &= 5x \\ 8x &= 5 \\ x &= \frac{5}{8} \end{aligned}$$

Test values:

- $x = -\frac{5}{2}$, $5x$ becomes negative, whilst the left is positive. Hence invalid solution.
- $x = \frac{5}{8}$:

$$\begin{aligned} \left| 3\left(\frac{5}{8}\right) - 5 \right| &= \left| \frac{15}{8} - \frac{40}{8} \right| \\ &= \left| -\frac{25}{8} \right| \\ &= \frac{25}{8} \\ 5\left(\frac{5}{8}\right) &= \frac{25}{8} \end{aligned}$$

Hence $x = \frac{5}{8}$ is the only solution.

(e) (3 marks)

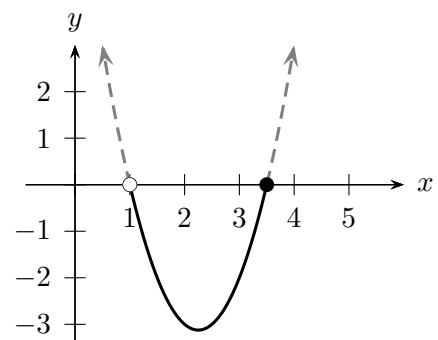
$$\frac{5}{x-1} \geq \frac{2}{x(x-1)^2} \quad x \neq 1$$

$$5(x-1) \geq 2(x-1)^2$$

$$2(x-1)^2 - 5(x-1) \leq 0$$

$$(x-1)(2(x-1) - 5) \leq 0$$

$$(x-1)(2x-7) \leq 0$$



$$\therefore 1 < x \leq \frac{7}{2}$$

Question 3 (Ziaziaris)

(e) (2 marks)

(a) (1 mark)

$$\begin{aligned} 180^\circ &\leftrightarrow \frac{\pi}{5} \\ \therefore 36^\circ &\leftrightarrow \frac{\pi}{5} \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} \cot \frac{11\pi}{6} &= \cot \left(2\pi - \frac{\pi}{6} \right) \\ &= -\cot \frac{\pi}{6} \\ &= -\frac{1}{\frac{1}{\sqrt{3}}} \\ &= -\sqrt{3} \end{aligned}$$

(c) i. (1 mark)

$$\begin{aligned} &\cos(x+y)\cos y + \sin(x+y)\sin y \\ &= \cos[(x+y)-y] \\ &= \cos x \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} &\cos^4 \theta - \sin^4 \theta \\ &= (\cancel{\cos^2 \theta + \sin^2 \theta}) \underbrace{(\cos^2 \theta - \sin^2 \theta)}_{=\cos 2\theta} \\ &= \cos 2\theta \end{aligned}$$

(d) (3 marks)

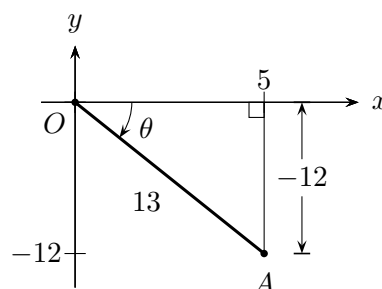
$$\begin{aligned} &\frac{\cos \theta}{1 - \sin \theta} \times \frac{\cos \theta}{\cos \theta} + \frac{1 - \sin \theta}{\cos \theta} \times \frac{(1 - \sin \theta)}{(1 - \sin \theta)} \\ &= \frac{\cos^2 \theta + (1 - \sin \theta)^2}{\cos \theta (1 - \sin \theta)} \\ &= \frac{\cos^2 \theta + 1 - 2\sin \theta + \sin^2 \theta}{\cos \theta (1 - \sin \theta)} \\ &= \frac{2 - 2\sin \theta}{\cos \theta (1 - \sin \theta)} \\ &= \frac{\cancel{2(1 - \sin \theta)}}{\cos \theta \cancel{(1 - \sin \theta)}} \\ &= 2 \sec \theta \end{aligned}$$

$$\tan 22.5^\circ = \sqrt{2} - 1$$

$$\begin{aligned} \tan 67.5^\circ &= \tan (22.5^\circ + 45^\circ) \\ &= \frac{\tan 22.5^\circ + \tan 45^\circ}{1 - \tan 22.5^\circ \tan 45^\circ} \\ &= \frac{(\sqrt{2} - 1) + 1}{1 - (\sqrt{2} - 1)} \\ &= \frac{\sqrt{2}}{2 - \sqrt{2}} \times \frac{2 + \sqrt{2}}{2 + \sqrt{2}} \\ &= \frac{2\sqrt{2} + 2}{4 - 2} \\ &= 1 + \sqrt{2} \end{aligned}$$

Question 4 (Lam)

(a) i. (2 marks)



$$\sin \theta = -\frac{12}{13}$$

ii. (2 marks)

$$\begin{aligned} \cos \theta &= \frac{5}{13} \\ \cos 2\theta &= 2\cos^2 \theta - 1 \\ &= 2\left(\frac{5}{13}\right)^2 - 1 \\ &= 2 \times \frac{25}{169} - 1 \\ &= \frac{50}{169} - \frac{169}{169} \\ &= -\frac{119}{169} \end{aligned}$$

(b) i. (2 marks)

$$\begin{aligned} \cos \theta &= -\frac{\sqrt{3}}{2} \\ \theta &= \pm 150^\circ \end{aligned}$$

ii. (3 marks)

$$\sqrt{2} \sin 3x = -1 \quad 0^\circ \leq x \leq 360^\circ$$

$$\sin 3x = -\frac{1}{\sqrt{2}} \quad 0^\circ \leq 3x \leq 1080^\circ \quad (c) \quad i. \quad (2 \text{ marks})$$

$$3x = \underbrace{225^\circ, 315^\circ, 585^\circ, 675^\circ, 945^\circ, 1035^\circ}_{\div 3}$$

$$x = 75^\circ, 105^\circ, 195^\circ, 225^\circ, 315^\circ, 345^\circ$$

iii. (3 marks)

$$2 \sin^2 \theta + 3 \cos \theta = 3$$

$$2(1 - \cos^2 \theta) + 3 \cos \theta = 3$$

$$2 - 2 \cos^2 \theta + 3 \cos \theta = 3$$

$$2 \cos^2 \theta - 3 \cos \theta + 1 = 0$$

Let $m = \cos \theta$,

$$2m^2 - 3m + 1 = 0$$

$$(2m - 1)(m - 1) = 0$$

$$m = 1, \frac{1}{2}$$

$$\therefore \cos \theta = 1, \frac{1}{2}$$

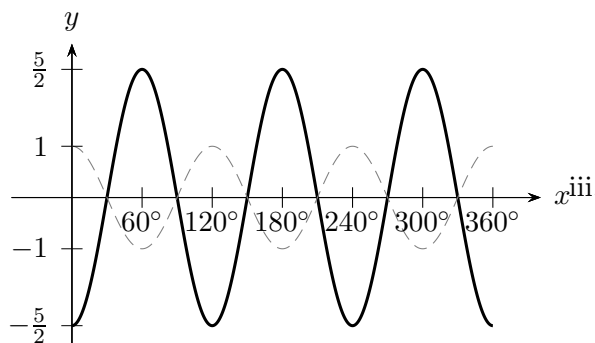
$$\theta = 0^\circ, 60^\circ, 300^\circ, 360^\circ$$

Question 5 (Lin)

(a) i. (2 marks)

- $T = \frac{360^\circ}{3} = 120^\circ$
- $a = \frac{5}{2}$

ii. (3 marks)



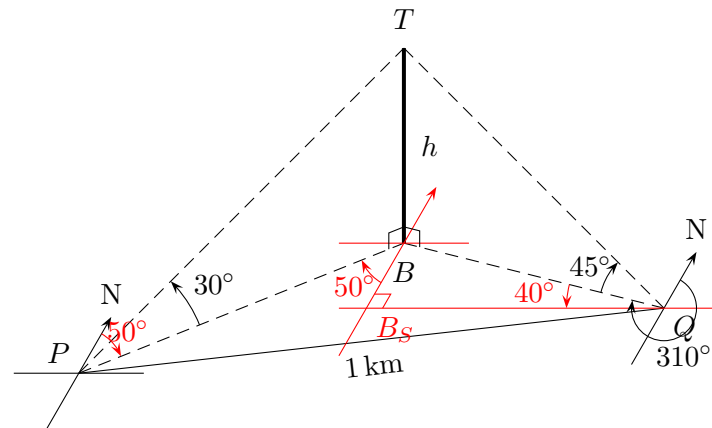
(b) (2 marks)

$$\frac{\sin \theta}{5} = \frac{\sin 28^\circ}{4}$$

$$\sin \theta = \frac{5 \sin 28^\circ}{4}$$

$$\theta = 35^\circ 56', 144^\circ 4'$$

(Ambiguous case)



- $\angle PBB_S = 50^\circ$
(alternate $\angle$, $NP \parallel BB_S$)
- $\angle BQB_S = 310^\circ - 270^\circ = 40^\circ$
- $\angle BB_SQ = 90^\circ$
- Hence

$$\angle QBB_S = 90^\circ - 40^\circ = 50^\circ$$

(complementary)

- Hence $\angle PBQ = 50^\circ + 50^\circ = 100^\circ$

ii. (1 mark)

$$\frac{h}{PB} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\therefore PB = h\sqrt{3}$$

In $\triangle TBQ$,

$$\frac{h}{QB} = \tan 45^\circ = 1$$

$$\therefore QB = h$$

iii. (2 marks)

Applying the cosine rule in $\triangle PBQ$, where $PQ = 1\,000$ m,

$$1\,000^2 = PB^2 + QB^2 - 2(PB)(QB) \cos 100^\circ$$

$$1\,000^2 = (h\sqrt{3})^2 + h^2 - 2(h\sqrt{3})(h) \cos 100^\circ$$

Factorising h^2 ,

$$1\,000^2 = h^2 (3 + 1 - 2\sqrt{3} \cos 100^\circ)$$

$$\therefore h^2 = \frac{1\,000^2}{4 - 2\sqrt{3} \cos 100^\circ}$$

iv. (1 mark)

$$h = 466 \text{ m}$$

v. (2 marks)

$$\begin{aligned} PB &= h\sqrt{3} \quad QB = h \\ A &= \frac{1}{2}ab \sin C \\ &= \frac{1}{2} \times h\sqrt{3} \times h \times \sin 100^\circ \\ &= \frac{1}{2} \times \frac{1\,000^2 \times \sqrt{3} \sin 100^\circ}{4 - 2\sqrt{3} \cos 100^\circ} \\ &= 185\,344 \text{ m}^2 \end{aligned}$$

Equating,

$$\begin{aligned} 2x &= 1 - \frac{3}{x+2} \\ 2x(x+2) &= (x+2) - 3 \\ 2x^2 + 4x &= x - 1 \\ 2x^2 + 3x + 1 &= 0 \\ (2x+1)(x+1) &= 0 \\ x &= -\frac{1}{2}, -1 \\ y &= -1, -2 \end{aligned}$$

(c) (3 marks)

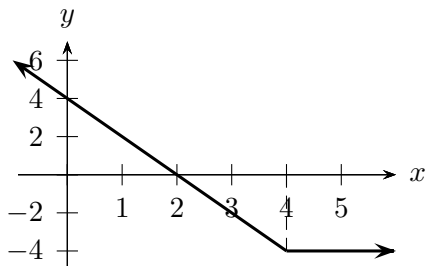
Question 6 (Zuber)

(a) (3 marks)

$$y = |-(x-4)| - x$$

Examine $|-(x-4)|$:

$$\begin{aligned} |-(x-4)| &= |x-4| \\ &= \begin{cases} (x-4) & (x-4) \geq 0 \\ -(x-4) & (x-4) < 0 \end{cases} \\ &= \begin{cases} (x-4) & x \geq 4 \\ -(x-4) & x < 4 \end{cases} \\ \therefore y &= \begin{cases} (x-4) - x & x \geq 4 \\ -(x-4) - x & x < 4 \end{cases} \\ &= \begin{cases} -4 & x \geq 4 \\ -2x + 4 & x < 4 \end{cases} \end{aligned}$$

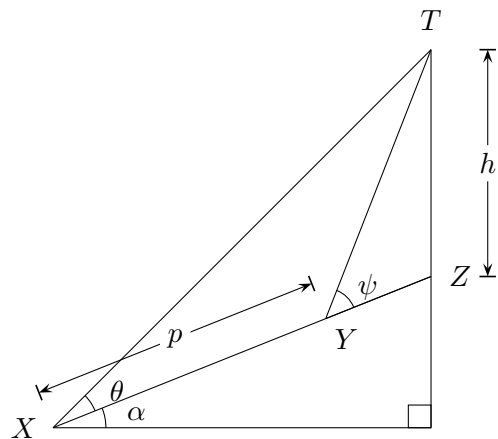


(b) (3 marks)

$$\begin{cases} y = 1 - \frac{3}{x+2} \\ y = 2x \end{cases}$$

$$\begin{aligned} \sin x - \sin 2x &= 0 \\ \sin x - 2 \sin x \cos x &= 0 \\ \sin x (1 - 2 \cos x) &= 0 \\ \sin x &= 0 \quad \cos x = \frac{1}{2} \\ \therefore x &= 0, \pi, 2\pi, \quad \frac{\pi}{3}, \frac{5\pi}{3} \end{aligned}$$

(d) i. (1 mark)



In $\triangle TYX$, $\angle XTY = \psi - \theta$.
Applying the sine rule,

$$\begin{aligned} \frac{TY}{\sin \theta} &= \frac{p}{\sin(\psi - \theta)} \\ \therefore TY &= \frac{p \sin \theta}{\sin(\psi - \theta)} \end{aligned}$$

- ii. (2 marks)
In $\triangle TYZ$,

$$\begin{aligned}
 \angle TZY &= \pi - \left(\frac{\pi}{2} - \alpha\right) \\
 &= \frac{\pi}{2} + \alpha \\
 \frac{h}{\sin \psi} &= \frac{TY}{\sin \left(\frac{\pi}{2} + \alpha\right)} \\
 \therefore h &= \frac{TY \sin \psi}{\sin \left(\frac{\pi}{2} + \alpha\right)} \\
 &= \frac{p \sin \theta \sin \psi}{\sin(\psi - \theta) \sin \left(\frac{\pi}{2} + \alpha\right)} \\
 &= \frac{p \sin \theta \sin \psi}{\sin(\psi - \theta) \left(\sin \frac{\pi}{2} \cos \alpha + \cos \frac{\pi}{2} \sin \alpha\right)} \\
 &= \frac{p \sin \theta \sin \psi}{\sin(\psi - \theta) \cos \alpha}
 \end{aligned}$$

- iii. (2 marks)

$$\begin{aligned}
 \alpha = \theta &= \frac{\pi}{6}, \psi = \frac{\pi}{4} \\
 h &= \frac{p \sin \frac{\pi}{6} \sin \frac{\pi}{4}}{\sin \left(\frac{\pi}{4} - \frac{\pi}{6}\right) \cos \frac{\pi}{6}} \\
 &= \frac{p \times \frac{1}{2} \times \frac{1}{\sqrt{2}}}{\left(\sin \frac{\pi}{4} \cos \frac{\pi}{6} - \cos \frac{\pi}{4} \sin \frac{\pi}{6}\right) \frac{\sqrt{3}}{2}} \\
 &= \frac{\frac{p}{2\sqrt{2}}}{\frac{\sqrt{3}}{2} \times \left(\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}\right)} \\
 &= \frac{\frac{p}{2\sqrt{2}}}{\frac{3}{4\sqrt{2}} - \frac{\sqrt{3}}{4\sqrt{2}}} \\
 &= \frac{2p}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}} \\
 &= \frac{2p(3 + \sqrt{3})}{9 - 3} \\
 &= \frac{p(3 + \sqrt{3})}{3}
 \end{aligned}$$